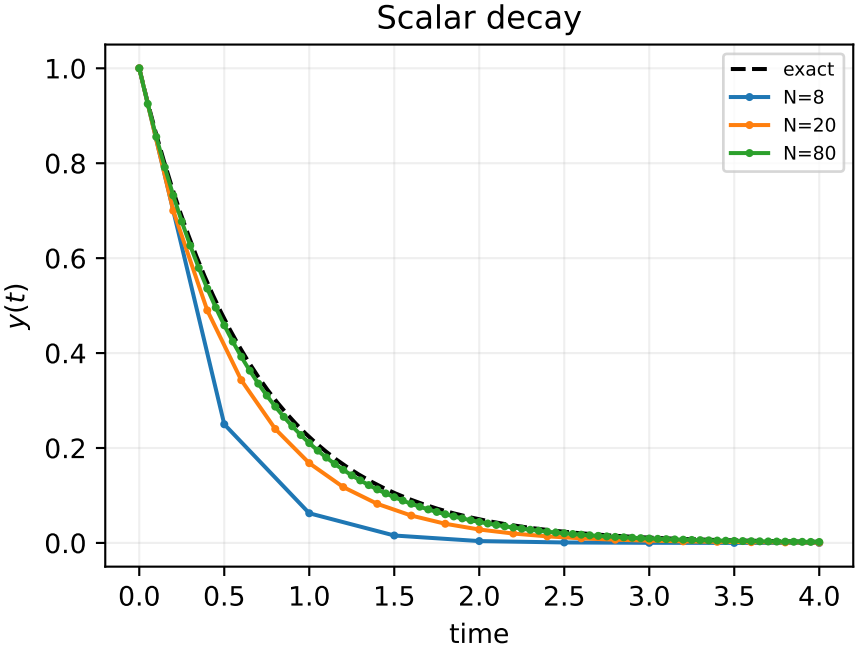
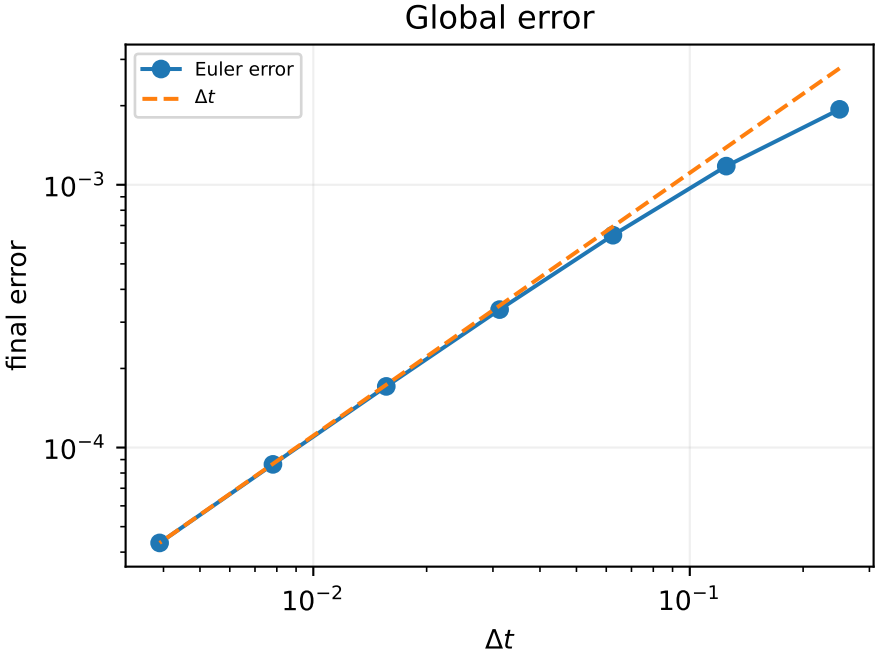
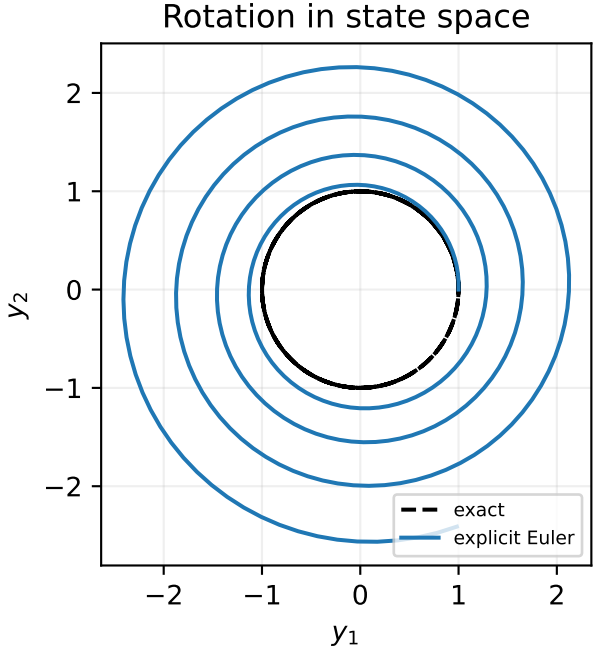
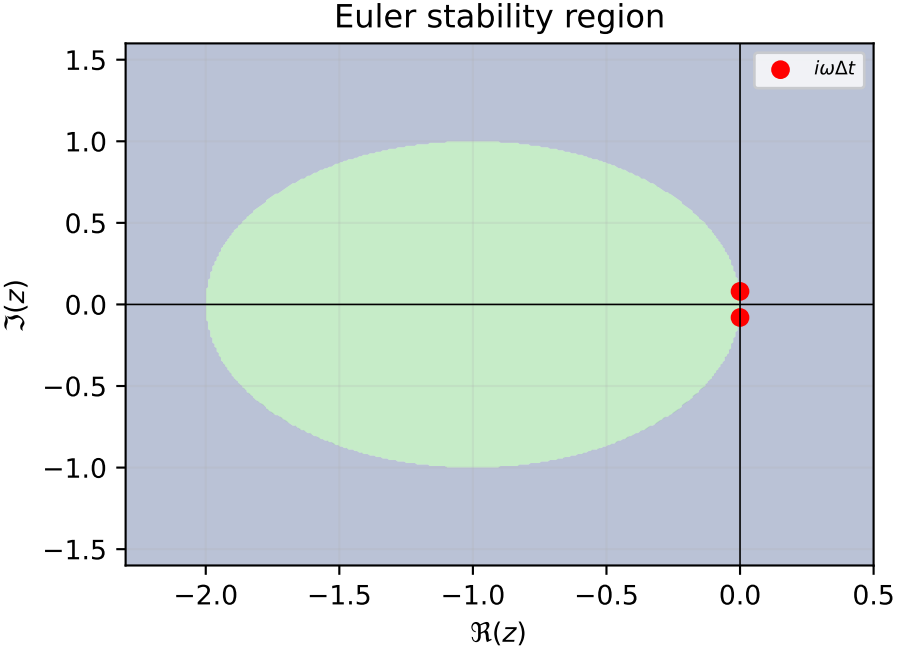

Scalar decay
exact
N=8
N=20
N=80
y(t)
time
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0


Global error
Euler error
Δt
final error
Δt
10^{-3}
10^{-4}
10^{-2}
10^{-1}


Rotation in state space
exact
explicit Euler
y_2
y_1
2
1
0
−1
−2
−2
−1
0
1
2


Euler stability region
$i\omega\Delta t$
$\Im(z)$
$\Re(z)$
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−1.5
−1.0
−0.5
0.0
0.5